

Performance Testing

Date	03 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Developer Tools
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Crop Prediction API (/predict)
Test Environment	Local Machine (Windows 11, Flask Server)
Test Date	02-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single crop prediction request	1	30	Prediction generated successfully
2	Multiple prediction requests	10	60	All requests processed successfully
3	Invalid input testing	5	30	Validation error displayed
4	Continuous prediction requests	20	120	Stable performance without crashes

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.72 sec	pass	Fast prediction
2	Response Time (Max)	< 5 seconds	1.84 sec	pass	Within acceptable limit
3	Throughput (Req/sec)	>10	18 req/sec	pass	Good throughput
4	Error Rate	< 1%	0%	pass	No request failures
5	CPU Utilization	< 80%	43%	pass	Stable CPU usage
6	Memory Utilization	< 80%	36%	pass	Efficient memory usage

Step 4: Observations & Analysis


OptiCrop AI
 Smart Agricultural Production Optimization Engine
 Predict the most suitable crop using Machine Learning and soil nutrient analysis.

Crop Recommendation

Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature (°C)

Humidity (%)

pH Value

Rainfall (mm)

AI Model Status

Model
Random Forest

Accuracy
99.31%

Dataset
2200 Records

Features
7 Parameters

Recommendation Result

kidneybeans

Algorithm
Random Forest Classifier

Prediction Type
Multi-Class Classification